

Amperova sila med dvema el. vodnikoma

$$\vec{F} = \frac{\mu_0 I_1 I_2}{4\pi} \int_{C_1} \int_{C_2} \frac{d\vec{l}_1 \times (d\vec{l}_2 \times (\vec{r}(l_2) - \vec{r}(l_1)))}{|\vec{r}(l_2) - \vec{r}(l_1)|^3} = \int_{C_1} I_1 d\vec{l}_1 \times \underbrace{\left[\frac{\mu_0 I_2}{4\pi} \int_{C_2} \frac{d\vec{l}_2 \times (\vec{r}(l_1) - \vec{r}(l_2))}{|\vec{r}(l_1) - \vec{r}(l_2)|^3} \right]}_{\vec{B} \text{ (gostota magh. polja)}}$$

Biot-Savartov zakon:

$$\vec{B}(\vec{r}) = \frac{\mu_0 I_2}{4\pi} \int_{C_2} \frac{d\vec{l}_2 \times (\vec{r} - \vec{r}(l_2))}{|\vec{r} - \vec{r}(l_2)|^3}$$

E.L. tok: $I = \frac{dq}{dt}$ [A]; v magnetostatiki je tok časovno konst.

Magnetne silnice:



Magnetna cirkulacija:

$$\Gamma_m = \oint_{C \cdot S} \vec{B} \cdot d\vec{r} = \oint_{C \cdot S} (\nabla \times \vec{B}) \cdot d\vec{S} \neq 0 \Rightarrow \nabla \times \vec{B} \neq 0 \Rightarrow \text{magh. polje je vrtinčno}$$

Magnetni pretok:

$$\Phi_m = \int_S \vec{B} \cdot d\vec{S} \quad \text{velja: } \oint_S \vec{B} \cdot d\vec{S} = 0, \text{ ker ni magnetnih monopolov}$$

$$\oint_{S=\partial V} \vec{B} \cdot d\vec{S} = \int_V \nabla \cdot \vec{B} dV \Rightarrow \nabla \cdot \vec{B} = 0 \Rightarrow \text{magh. polje je brezizvorno}$$

Gostota el. toka: $I = \int_S \vec{j} \cdot d\vec{S}$ (presek žice) ($\vec{j}$ ni nujno vezan na vodnike)

Amperov izrek:

$$\begin{aligned} \Gamma_m &= \oint_C \vec{B} \cdot d\vec{l} = \oint_C \left[\frac{\mu_0 I}{4\pi} \int_{C'} \frac{d\vec{l}' \times (\vec{r}(l) - \vec{r}(l'))}{|\vec{r}(l) - \vec{r}(l')|^3} \right] \cdot d\vec{l} = \iint_{C \cdot C'} \frac{\mu_0 I}{4\pi} \left[d\vec{l}' \times \frac{\vec{r}(l) - \vec{r}(l')}{|\vec{r}(l) - \vec{r}(l')|^3} \right] \cdot d\vec{l} = \\ &= \oint_{C'} \underbrace{\left[\frac{\mu_0 I}{4\pi} (d\vec{l}' \cdot d\vec{l}) \right]}_{dS} \cdot \underbrace{\left[\frac{(\vec{r}(l) - \vec{r}(l'))}{|\vec{r}(l) - \vec{r}(l')|^3} \right]}_{= \oint_S d\Omega = 4\pi} = \mu_0 I \end{aligned}$$

magh. cirk. je enaka zaobjemu el. toku.

$$\oint_{\partial S} \vec{B} \cdot d\vec{r} = \mu_0 I = \mu_0 \int_S \vec{j} \cdot d\vec{S}$$

$$\oint_{\partial S} \vec{B} \cdot d\vec{r} \Rightarrow \nabla \times \vec{B} = \mu_0 \vec{j} \quad \text{Dif. oblika Amp. izreka}$$

Magnetni potencial: $\vec{B} = \nabla \times \vec{A} \rightarrow \Phi_m = \int_S \vec{B} \cdot d\vec{S} = \int_S (\nabla \times \vec{A}) \cdot d\vec{S} = \oint_{\partial S} \vec{A} \cdot d\vec{r}$

Vekt. potencial tuljave:

• znotraj: $\vec{B} = (0, 0, B_z) \rightarrow \vec{A} = \frac{1}{2} \vec{B}_z \times \vec{r}$ (se izkaže)

• zunaj: $\vec{B} = 0 \rightarrow \vec{A}_{\text{zunaj}} \neq 0$ (prek magh. pretoka) $\rightarrow \vec{A} = C \cdot \vec{B}_z \times \frac{\vec{r}}{r^2}$

• določimo C, da je $\vec{A}$ na meji zvezen: $C = \frac{a^2}{2}$ (polmer tuljave)

• umeritev $\vec{A}$: uvedemo $\vec{A}' = \vec{A} - \nabla \chi(\vec{r})$; $\vec{B}' = \vec{B} - \nabla \chi(\nabla \chi) = 0$

• npr. $\chi = \frac{B_z a^2}{2} \arctan \frac{y}{x} \Rightarrow$ zanj tuljave velja $\vec{A} = \frac{a^2}{2} \vec{B}_z \times \frac{\vec{r}}{r^2} - \nabla \left(-\frac{B_z a^2}{2} \arctan \frac{y}{x} \right) = \frac{B_z a^2}{2} \frac{2\pi}{a} \delta(\phi - \pi) \hat{e}_\phi \rightarrow \vec{A} = 0$ skrajni povesod (razen pri $\phi = \pi$)

Kirchoffova enačba

• Ampere: $\mu_0 \vec{j} = \nabla \times \vec{B} = \nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}$

• Helmholtzov izrek: vsako vekt. polje lahko zapišemo kot $\vec{A} = \vec{A}_1 + \vec{A}_2$; $\nabla \cdot \vec{A}_1 = 0$, $\nabla \times \vec{A}_2 = 0$.

• ker $\vec{B} = \nabla \times \vec{A} = \nabla \times \vec{A}_1 + \nabla \times \vec{A}_2 = \nabla \times \vec{A}_1$, lahko izberemo $\vec{A}_2 = 0 \Rightarrow \nabla \cdot \vec{A} = \nabla \cdot \vec{A}_1 = 0$

$$\mu_0 \vec{j} = -\nabla^2 \vec{A}$$

• iz Poissonove en. sklepamo rešitev: $\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int_V \frac{\vec{j}(\vec{r}')}{|\vec{r} - \vec{r}'|} d^3 r'$

• če na rešitev delujemo z rotorjem, dobimo Biot-Savartov zakon

Magnetna energija

• v zun. polju (premaknemo zanko): $\vec{F} = \int_C I d\vec{l} \times \vec{B} = I \int_C (\vec{r} \times \vec{B}) d\vec{l}$

$$\rightarrow dA = -\vec{F} \cdot d\vec{r} = -I \int_C [\vec{r} \times \vec{B}] \cdot d\vec{r} = I B dS \rightarrow A = -I \Phi_m$$

$$\rightarrow A = -I \int_S \vec{B} \cdot d\vec{S} = \int_S (\nabla \times \vec{A}) \cdot d\vec{S} = -I \int_S \vec{A} \cdot d\vec{r} + I \int_{C_1} \vec{A} \cdot d\vec{r}$$

$$\rightarrow \text{posled. za } \vec{j}: A = - \int_V \vec{j} \cdot \vec{A} d^3 r + \int_{C_1} \vec{A} \cdot d\vec{r}$$

2 (po pravilu) 1 (pred pravilu)

$$\rightarrow W = - \int_V \vec{j} \cdot \vec{A} d^3 r$$

Mag. energija kot funkcional toka

$$\text{• Kirchoff: } W = - \frac{\mu_0}{4\pi} \int_V \frac{\vec{j}(\vec{r}) \cdot \vec{j}(\vec{r}')}{|\vec{r} - \vec{r}'|} d^3 r d^3 r'$$



elotna mag. eh.

$$dW = - \int d\vec{j} \cdot \vec{A} \, d^3\vec{r} = - \int_0^1 \underbrace{d\omega}_{\omega} \underbrace{\vec{\alpha}}_{\vec{A}} \, d^3\vec{r} \rightarrow \boxed{W = \frac{1}{2} \int \vec{j} \cdot \vec{A} \, d^3\vec{r}}$$

• energija NE upoštevna dela, potrebnega za vzpostavitev $\vec{J}$: $P = -UI = -I \int \vec{E} \cdot d\vec{r} = I \frac{d}{dt} \int \vec{B} \cdot d\vec{S} = \frac{dW}{dt}$

$$\Rightarrow W = \frac{1}{2\mu_0} \int_V B^2 d^3r$$
$$F = \int_V \mathbf{j} \times \vec{B}_{\text{ext}} d^3\vec{r} = \frac{1}{\mu_0} \int_V (\nabla \times \vec{B}_{\text{ext}}) \times \vec{B}_{\text{ext}} d^3\vec{r} = \frac{1}{\mu_0} \int_V (\nabla \times \vec{B}_{\text{ext}}) \times (\vec{B}_{\text{ext}}) d^3\vec{r} = \frac{1}{\mu_0} \int_V (\nabla \times \vec{B}_{\text{ext}}) \times \vec{B}_{\text{ext}} d^3\vec{r} = \frac{1}{\mu_0} \int_V \nabla \cdot (\vec{B}_{\text{ext}} \otimes \vec{B}_{\text{ext}}) - \frac{1}{2} \nabla B^2 d^3\vec{r} = \frac{1}{\mu_0} \int_V (\vec{B}_{\text{ext}} \otimes \vec{B}_{\text{ext}} - \frac{1}{2} B^2 \mathbf{I}) d^3\vec{r}$$
$$\vec{F}_i = \oint_{\partial V} T_{ik} n_k dS \quad \leftarrow \text{normale na površine}$$

$$T_{ik} = \frac{1}{\mu_0} \left[B_i B_k - \frac{1}{2} B^2 \delta_{ik} \right]$$

$$\vec{A}(\vec{r}) = \frac{1}{4\pi} \int \frac{(\vec{r}-\vec{r}')}{|\vec{r}-\vec{r}'|^3} d^3r' = \frac{1}{4\pi} \int \frac{(\vec{r}-\vec{r}')}{r^3} d^3r' + \frac{M_0}{4\pi} \int \frac{(\vec{r}') \cdot (\vec{r}-\vec{r}')}{r^3} d^3r' = \frac{M_0}{4\pi r} \underbrace{\int (\vec{r}') d^3r'}_{\text{monopol vector}} + \underbrace{\frac{M_0}{4\pi r^3} \int (\vec{r}') \cdot (\vec{r}-\vec{r}') d^3r'}_{\text{dipol tensor}} + \dots \approx \frac{M_0}{4\pi r} + \frac{M_0}{4\pi r^3} \vec{r} \cdot \vec{r}' + \dots$$

$$\vec{m} = \frac{1}{2} \int \vec{r}' \times \vec{j}(\vec{r}') d^3\vec{r}'$$

mag. polje mag. dipola

$$\vec{B} = \nabla \times \vec{A} = \nabla \times \left(\frac{\mu_0}{4\pi} \frac{\vec{m} \times \vec{r}}{r^3} \right) = \frac{\mu_0}{4\pi} \frac{r^2 \vec{m} + 3\vec{r}(\vec{m} \cdot \vec{r})}{r^5}$$

$$\oint_V \vec{F} \cdot \vec{x} \nabla(\vec{r}') d^3\vec{r}' = \frac{1}{2} \int_V \vec{x} \cdot (\vec{t}) dV = \frac{1}{2} \hat{e}_z \int_{2\pi a}^1 \frac{1}{5} \pi a^2 \hat{e}_z$$
$$= - \int_V \vec{A} \cdot d\vec{r} = - \int_V \vec{A}(\vec{r}_c) \cdot d\vec{r} = - \int_V (\vec{r} - \vec{r}_c) \cdot \nabla_c A(\vec{r}_c) \cdot d\vec{r} = - A(\vec{r}_c) \cdot \int_V d\vec{r} = - \frac{1}{\epsilon_0} \frac{1}{4\pi} \int_V \nabla_c \cdot \left(\frac{\vec{r} - \vec{r}_c}{|\vec{r} - \vec{r}_c|^3} \right) \cdot \nabla_c A(\vec{r}_c) \cdot d\vec{r}$$

$$\left. \begin{aligned} \int (j_i(\vec{r}-\vec{r}_i)) d^3r &= - \int (j_i(\vec{r}-\vec{r}_i)) d^3r \\ \text{li = simetrija} \end{aligned} \right\} \Rightarrow W = - \frac{\partial A_i(\vec{r}_i)}{\partial r_i} \frac{1}{2} \int (j_i(\vec{r}-\vec{r}_i) - j_i(\vec{r}-\vec{r}_i)) d^3r = - \frac{1}{2} \int ((\vec{r}-\vec{r}_i) \cdot \nabla) \vec{A}(\vec{r}_i) - (\vec{r}-\vec{r}_i) \cdot \nabla A(\vec{r}_i) d^3r = - \frac{1}{2} \left(\int (\vec{r}-\vec{r}_i) d^3r \right) \cdot [\nabla \times \vec{A}(\vec{r}_i)] =$$

$$dW = \vec{F} \cdot d\vec{r}$$
$$W = -\vec{M} \cdot d\vec{\phi}$$

$$(\vec{m} \cdot \vec{B}) = d\vec{m} \cdot \vec{B} = (d\vec{\phi} \times \vec{m}) \cdot \vec{B} = d\vec{\phi} \cdot (\vec{m} \times \vec{B}) \Rightarrow \boxed{\vec{M} = \vec{m} \times \vec{B}}$$

klasična formulacija: $\oint_{\partial V} \vec{A} \cdot d\vec{S} = \int_V \nabla \cdot \vec{A} d^3\vec{r}$
 koristimo izrek G.O. za $\vec{r} \otimes \vec{r} \otimes \vec{A}$: $\oint_{\partial V} r_j A_i n_k dS = \int_V (r_j A_i + r_i A_j + r_j \nabla \cdot A) d^3\vec{r}$

porabimo za $\vec{A} = \vec{j}$: $\oint_{\partial V} \vec{r} \cdot \vec{j} \, d\vec{s} = \int_V (\vec{r} \cdot \vec{j})_{,j} \, d\vec{r} \Rightarrow \boxed{\int_V \vec{r} \cdot \vec{j} \, d\vec{r} = -\int_V r_j \, d\vec{r}}$
(računamo vse
težavne zanke)